

```
#include <stdio.h>

typedef struct
{
    char id[5];
    int weight;
} Package;

int comparisons = 0;

void merge(Package a[], int low, int mid, int high)
{
    Package temp[50];

    int i = low;
    int j = mid + 1;
    int k = low;

    while (i <= mid && j <= high)
    {
        comparisons++;

        /* <= preserves the original order of equal weights */
        if (a[i].weight <= a[j].weight)
        {
```

```
        temp[k] = a[i];  
        i++;  
    }  
    else  
    {  
        temp[k] = a[j];  
        j++;  
    }  
  
    k++;  
}
```

```
while (i <= mid)  
{  
    temp[k] = a[i];  
    i++;  
    k++;  
}
```

```
while (j <= high)  
{  
    temp[k] = a[j];  
    j++;  
    k++;  
}
```

```

    }

    for (i = low; i <= high; i++)
    {
        a[i] = temp[i];
    }
}

void mergeSort(Package a[], int low, int high)
{
    if (low < high)
    {
        int mid = (low + high) / 2;

        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);

        merge(a, low, mid, high);
    }
}

```

```

void display(Package a[], int n)
{
    int i;

```

```
printf("Package ID\tWeight\n");
```

```
for (i = 0; i < n; i++)
```

```
{
```

```
    printf("%s\t\t%d\n", a[i].id, a[i].weight);
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
    Package a[] =
```

```
{
```

```
    {"P1", 20},
```

```
    {"P2", 15},
```

```
    {"P3", 20},
```

```
    {"P4", 10},
```

```
    {"P5", 15},
```

```
    {"P6", 20},
```

```
    {"P7", 25},
```

```
    {"P8", 10}
```

```
};
```

```
int n = 8;
```

```
printf("Original Packages:\n");  
display(a, n);  
  
mergeSort(a, 0, n - 1);  
  
printf("\nAfter Merge Sort:\n");  
display(a, n);  
  
printf("\nNumber of comparisons = %d\n", comparisons);  
  
return 0;  
}
```